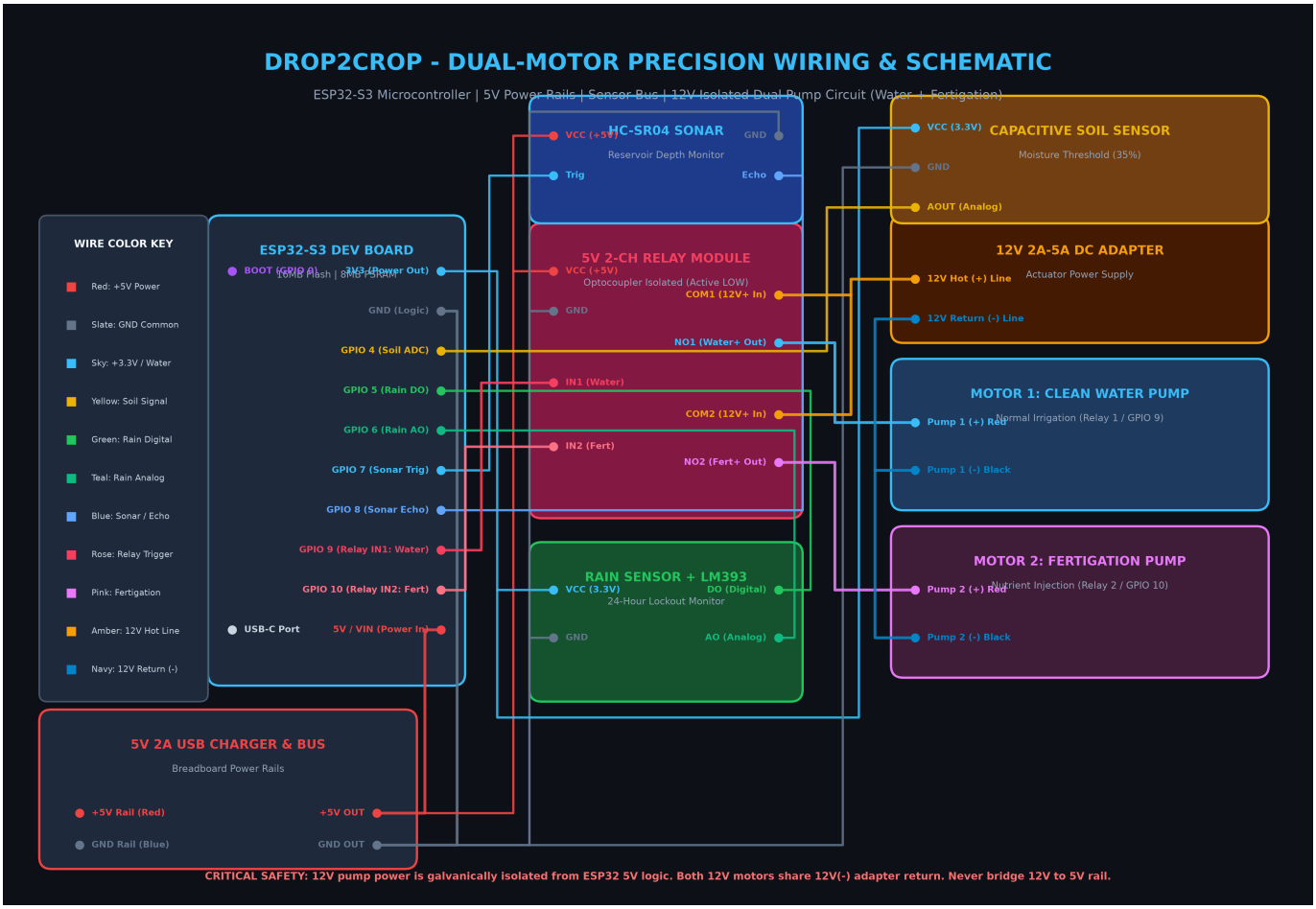
Drop2Crop — Hardware Wiring Diagram & Manual

Dual-Motor Electrical Pinout, Power Bus Distribution & Relay Circuitry

1. System Master Wiring Schematic (Dual 12V Motors)

The schematic below details all wiring connections between the ESP32-S3, 2-channel relay module, HC-SR04 sonar sensor, capacitive soil moisture sensor, rain detector, breadboard power rails, and the two independent 12V DC pump circuits (Watering & Fertigation).



2. Complete ESP32-S3 Pinout Connection Table

Module / Component	Component Pin	ESP32-S3 Pin / Bus	Voltage	Signal Description
Capacitive Soil Sensor	VCC	**ESP32 3V3 Pin**	3.3V DC	Regulated 3.3V supply (prevents ADC drift)
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	AOUT	**GPIO 4 (ADC1_CH3)**	0 - 3.3V	Analog soil moisture signal (1350-3150)
Rain Sensor (LM393)	VCC	**ESP32 3V3 Pin**	3.3V DC	Logic supply (safe for ESP32 inputs)
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	DO	**GPIO 5**	3.3V Logic	Digital rain trigger (LOW = Rain)
	AO	**GPIO 6 (ADC1_CH5)**	0 - 3.3V	Analog precipitation density
HC-SR04 Ultrasonic	VCC	**Breadboard +5V Rail**	5.0V DC	5V power for sonar transmitter
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	TRIG	**GPIO 7**	3.3V Logic	10µs ultrasonic trigger pulse
	ECHO	**GPIO 8**	Input	Echo pulse return width
5V 2-Channel Relay	VCC	**Breadboard +5V Rail**	5.0V DC	Relay coil power supply
	GND	**Breadboard GND Rail**	0V	Common Ground reference
	IN1	**GPIO 17 *(mirrored: 9)***	Active LOW	Relay 1 trigger: Motor 1 (Clean Water Pump)
	IN2	**GPIO 18 *(mirrored: 10)***	Active LOW	Relay 2 trigger: Motor 2 (Fertigation Pump)
Motor 1: Clean Water	(+) Red	**Relay 1 NO1**	+12V Switched	Passes through Relay 1 (COM1 connected to 12V+)
	(-) Black	**12V Adapter (-) Line**	0V (12V Ret)	Direct connection to 12V DC power return
Motor 2: Fertigation	(+) Red	**Relay 2 NO2**	+12V Switched	Passes through Relay 2 (COM2 connected to 12V+)
	(-) Black	**12V Adapter (-) Line**	0V (12V Ret)	Direct connection to 12V DC power return
Onboard Hardware	BOOT Button	**GPIO 0**	Active LOW	Hold 3s to trigger WiFi Captive Portal

3. Dual-Motor Power Distribution Architecture (5V Logic + 12V Actuator Isolation)

To guarantee absolute stability and protect the ESP32-S3 from voltage spikes caused by two high-power inductive motors:

- **5V Mobile Charger:** Powers ESP32-S3 via USB-C or VIN/GND pins, plus 5V breadboard rail for Relay coils and Sonar.
- **3.3V Logic Bus:** Derived from ESP32 internal LDO regulator; powers Soil and Rain sensors only.
- **12V Isolated High-Power Actuators:** The 12V Universal Adapter powers both motors completely isolated from the microcontroller. The 12V (+) hot wire bridges to Relay 1 **COM1** and Relay 2 **COM2**. Relay **NO1** routes to Motor 1 (+) and Relay **NO2** routes to Motor 2 (+). Both motors connect their (-) return wires directly to the 12V adapter (-) line.
- **Strict Electrical Rule: NEVER connect the 12V adapter (+) or (-) wires to the ESP32 pins or the 5V breadboard rails.**

ELECTRICAL SAFETY WARNING: Disconnect the 12V adapter from mains AC before wiring the relay terminal blocks. Verify all polarities before powering up. Both motors are optoisolated on the relay module.